

Injury severity prediction from two-vehicle crash mechanisms with machine learning and ensemble models

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Abstract—Machine learning algorithms aim to improve the power of predictors over conventional regression models. This study aims to tap the predictive potential of crash mechanism-related variables using ensemble machine learning models. The results demonstrate selected models can predict severity at a high level of accuracy. The stacking model with a linear blender is preferred for the designed ensemble combination. Most bagging, boosting, and stacking algorithms perform well, indicating ensemble models are capable of improving upon individual models.

Index Terms—Injury severity, machine learning algorithms, vehicle crashes, ensemble technique, crash mechanisms

I. INTRODUCTION

PREDICTING injury severity from automobile crashes is a necessary component of traffic safety studies. Efforts to reduce the collision impact received by vehicle occupants in car crashes rely on accurate predictions. Recent data mining and machine learning (ML) algorithms offer new insight for weighting and combining the factors to predict injury severity with relatively high accuracy [1], [2], [3], [4], [5], [6].

Traditionally statistical regression techniques have been used to explain the factors underlying severe injuries as well as to predict the severity outcome of specific crashes. The most commonly-used regression prediction methods are multi-level ordered logit models, binomial logistic models, and their variants. These regression models can directly provide a reasonable theoretical interpretation and precisely reveal the relationships of different factors through an analytical approach [7]. However, they impose specific mathematical forms between dependent and explanatory variables, which rely on several predefined assumptions [8]. Compared to other algorithms, the prediction performance of regression models is weak due to their fixed assumptions [9].

ML models are designed to improve the prediction accuracy with a non-parametric process [10]. Without underlying functions, ML provides powerful algorithms for predictions and classifications, but the problem is the difficulty of explaining the learning process [11]. This ‘black box’ tactic makes the algorithms weak for the conceptual explanation of models (interpretability) [12]. Sensitivity analysis is one of the methods to mitigate this ‘black box’ issue. [13], [14], [15], [16], [17], [18]

Ensemble learning combines the outputs of multiple statistical and machine learning models to obtain an improved

prediction estimate and range. Ensemble techniques have been widely deployed in many fields, including weather forecasting and medicine [19], [20], [21], [22], and they have also been conducted in recent crash injury analysis [16], [17].

Previous related studies found that collision speed, crash type, and passenger restraint are significant factors in predicting the injury severity, which are related to mechanisms in two-vehicle involved crashes [23], [24], [25], [26]. This paper aims to highlight the importance of the mechanism-related predictor based on kinetic energy (found in [27], [28]) and use ML and ensemble algorithms to evaluate injury severity in two-vehicle crashes.

The structure of this paper is organized as follows. In section II and section III, we introduce and specify the datasets and methods applied in this study. The results and discussions about the prediction outcomes are presented in section IV. Finally, section V summarizes and provides suggestions for further studies.

II. DATA AND VARIABLES DESCRIPTION

National Highway Traffic Safety Administration (NHTSA) provides a selection of police-reported crash data, the Crashworthiness Data System (CDS), which is a subset of the National Automotive Sampling System (NASS). It was designed to reveal the multidisciplinary relationship in automobile crash investigation through the cooperation of engineering and trauma centers [29]. It recorded a total of 49,639 crashes data between 2009 and 2015 (codes changed after 2008), mainly collecting information about injury details of occupants, and vehicle interior and exterior (environment) conditions.

For this study, we only consider two-vehicle crashes, the theory of which can be explained by a simplified crash mechanism. Since the dependent variable Maximum Abbreviated Injury Score (MAIS) is essential for each vehicle, we remove data records where this value was omitted, as shown in Figure 1. MAIS, which includes ordered classifications from 0 to 6+, which represents the severest occupant injury severity in this vehicle. Level 0 (Property Damage Only) and Level 1 (Minor Injury) constitutes 76.23% of the data, while Level 4 (Severe Injury) and Level 5 (Major Injury) account for small proportions (2.15% and 1.10% respectively).

The models in this paper estimate injury severity at the vehicle level, which means each observation represents a vehicle involved in crashes. The Newtonian mechanism explains how occupants in each vehicle receive severe injuries in crashes.

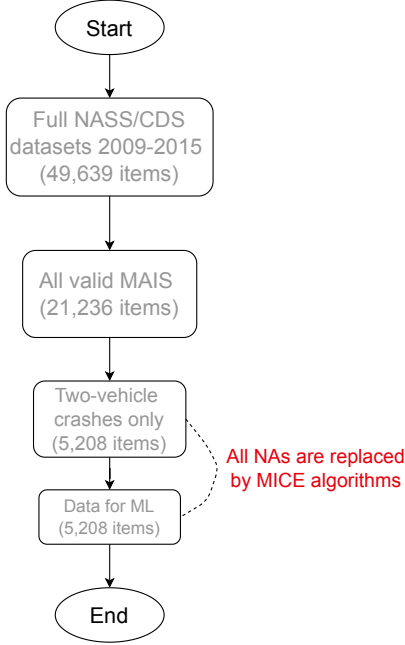


Figure 1: The preprocessing of data for ML algorithms

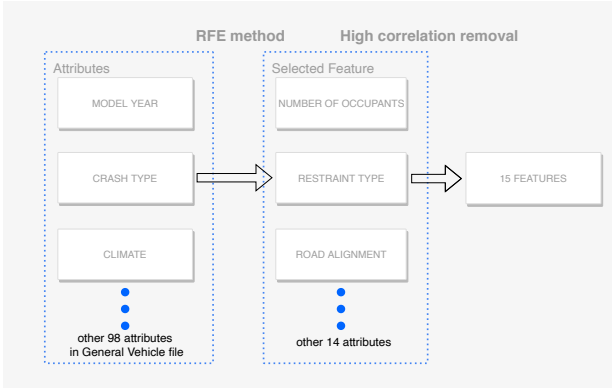


Figure 2: A graphical description of the feature selection process

The ratio of the acceleration (a) imposed on bodies of the subject (s) and the partner (p) vehicle is proportional to energy absorption (E), and is inversely proportional to the vehicle mass (m) and the crash deformation (d), as shown in Equation 1.

$$\frac{\bar{a}_s}{\bar{a}_p} = \frac{E_s}{E_p} \times \frac{\bar{m}_p}{\bar{m}_s} \times \frac{d_p}{d_s} \quad (1)$$

We also check other variables that may impact the crash, including both internal (general information of vehicles) and external (surrounding conditions) factors. To avoid the redundancy and over-fitting problems, we eliminate all highly-correlated attributes and select several top attributes by a Recursive Feature Elimination (RFE) method, as illustrated in Figure 2. RFE, like stepwise regression, is a backward selection method that estimates a model on all features, computes importance scores, and removes the least important ones [30]. Table I summarizes dependent and selected explanatory

variables as well as their descriptions. For simplicity, we hereafter use codes (in parentheses after variable names) to represent variables.

Note that all missing numerical data are generated by the Predictive Mean Matching (PMM) method. Missing categorized data is replaced through Logistic regression (for binary levels) and Bayesian polytomous regression (for multiple levels) to create Multiple Imputations by Chain Equations (MICE) [31].

III. METHODOLOGY

First, we seek to find the most satisfactory single ML algorithms and ensemble models based on their performance. For that, we first divide the dataset into the ‘Training’ set (75%) and the ‘Testing’ set (25%) and train the models through an n -fold cross-validation by the ‘Training’ set. That means, during the training process, the data will be split into n subsets ($n - 1$ for training and 1 for validation), and the hyper parameters will be tuned by repeating the procedure n times. The dedicated ‘Testing’ set can then test the trained models. According to their independent prediction outcomes, every variable’s importance is computed by an ensemble sensitivity analysis. Finally, we can rank the importance of variables and analyze their magnitude of effects.

The R package ‘caret’ applies for all of the machine learning and ensemble algorithms and tunes the hyper parameters for model performance through the n -fold cross-validation technique (in this study $n = 10$).

A. Statistical models

1) *Proportional odds ordered logistic regression model (POLR)*: To compare with machine learning models, we first apply the POLR model as a traditional statistical method in the multi-class prediction. Alternative models should be explored in future studies.

Assume that a latent variable Y follows a linear relationship with explanatory variables X :

$$Y^* = \beta_0 + \beta X + \varepsilon \quad (2)$$

where β_0 is the intercept of this linear function, β is the corresponding coefficient of variable X , and ε is the error term in this function.

Then, the values of Y^* and coefficients and intercepts β are estimated through the Principle of Maximum Likelihood method in Equation 3. Assume the latent variable Y follows a link function $g(\mu)$.

$$L = \prod_{j=1}^N \prod_{i=1}^M (g(\beta_{0,j} - \beta X_i) - g(\beta_{0,j-1} - \beta X_i)) \quad (3)$$

in which:

N is the total number of independent variables;

M is the number of classifications for the dependent variable.

After the fitting process for Y and coefficients β , the predicted possibility of each MAIS level with the logit link

Variable	Variable Description	Freq.	Ratio	Total
Dependent variable				
MAIS	0 - Property Damage Only	1,796	34.49%	5,208
	1 - Minor Injury	2,174	41.74%	
	2 - Moderate Injury	536	10.29%	
	3 - Serious Injury	296	5.68%	
	4 - Severe Injury	112	2.15%	
	5 - Major Injury	57	1.10%	
	6 ⁺ - Fatality (Untreatable)	237	4.55%	
MAIS (binary)	0 - Not Serious Injury (3-)	4,506	86.52%	
	1 - Serious Injury (3 & 3+)	702	13.48%	
MAIS (ternary)	0 - Property Damage Only (0)	1,796	34.49%	
	1 - Minor Injury (1)	2,174	41.74%	
	2 - More Than Moderate Injury (2 & 2+)	1,238	23.77%	
Independent variables				
Internal influences				
Crash Type (ACCTYPE)	1 - Straight-line crashes	1,229	23.60%	
	2 - Side-impact crashes	387	7.43%	
	3 - Angle crashes	3,592	68.97%	
Collision Angles (ANGTHIS/ANGOTHER)	Collision angles for vehicles (0-360 degree)	-	-	
Energy Absorption (ENERGY)	Energy absorption by this vehicle (in kilojoules)	-	-	
Model Year (MODELYR)	The year in which this vehicle was manufactured	-	-	
Number of Occupants (OCUPANTS)	The number of occupants seated in this vehicle (1-8)	-	-	
Restrain Type (RESTYPE)	1 - Airbags front only	2,025	38.88%	
	2 - Airbags front & side	3,164	60.76%	
	3 - Passive belts or unknown	19	0.36%	
	/no airbags			
Vehicle Type (VEHTYPE)	1 - Trucks	2,198	42.20%	
	0 - Passenger cars	3,010	57.80%	
External influences				
Road Alignment (ALIGNMENT)	1 - Straight	4,370	83.91%	
	2 - Curve right	410	7.87%	
	3 - Curve left	428	8.22%	
Weather Conditions (CLIMATE)	1 - Clear	4,031	77.40%	
	0 - Not clear	1,177	22.60%	
Number of Lanes (LANE)	The number of lanes on the road (1-7)	-	-	
Light Conditions (LGTCND)	1 - Daylight	3,619	69.49%	
	0 - No daylight	1,589	30.51%	
Road Profile (PROFILE)	1 - Level	3,840	73.73%	
	2 - Uphill	666	12.79%	
	3 - Crest	44	0.85%	
	4 - Downhill	647	12.42%	
	5 - Sag	11	0.21%	
Surface Type (SURTYPE)	1 - Asphalt	4,659	89.46%	
	0 - Others	549	10.54%	
Trafficway Flow (TRAFFLOW)	0 - Not Divided	3,330	63.94%	
	1 - Divided/No Barrier	1,088	20.89%	
	2 - Divided with Barrier	451	8.66%	
	3 - One Way	339	6.51%	

Table I: Description of variables in models

function (or any other link functions) is computed in [Equation 4](#).

$$\begin{aligned}
 P(0 < Y \leq 1) &= \frac{1}{1 + \exp(-Y_1)} \\
 P(1 < Y \leq 2) &= \frac{1}{1 + \exp(-Y_2)} - \frac{1}{1 + \exp(-Y_1)} \\
 &\vdots \\
 P(5 < Y \leq 6+) &= 1 - \frac{1}{1 + \exp(-Y_5)}
 \end{aligned} \tag{4}$$

The bi-class model shares a similar form with the multi-class one and can be estimated by the Generalized Linear Model (GLM).

B. Machine learning models

ML tools for crash-caused injury severity prediction, can be summarized into several categories.

1) *k-nearest neighbors (kNN) classification*: As a representative model of instance-based learning, the k-nearest neighbors (kNN) algorithm forecasts the classifications by estimating Euclidean distances for continuous variables (or other metrics for categorized variables) between training instances and the generated instance within k closest observations [32], [33], [34]. In general cases, $k = 10$ is recommended to conduct the first trial. It belongs to lazy-learning algorithms that delay the training data generation process until the system receives queries. Thus, the kNN algorithm can estimate each new instance individually and locally rather than the whole instance space. Another benefit of the kNN algorithm is that classifiers cost little to update for new instances with known classifications. However, it requires extensive space to store the training data and compute the distances simultaneously, particularly for datasets with a large number of attributes. Due to its few parameters and simple implementation, this algorithm has been applied in many crash severity related studies to compare with other algorithms [9], [35], [36].

2) *Support vector machines (SVM)*: SVM is a supervised ML tool based on statistical theory and the structural risk minimization principle [37], [38], [39]. It aims to train the non-linear learning machines and exploit the generated optimization theory in high dimensional feature spaces. The main algorithm for this machine is to build several separating hyperplanes and find the optimal one to classify different groups. To apply this method into analysis and maximize its performance, two input values C (the cost in the radial basis kernel function) and γ (an error penalty factor that controls the minimization problem) should be first tuned during the training process. As a typical selection, the C-classification SVM integrates with a radial basis function performs well for discrete outcomes prediction [40]. The problem that occurred in multi-class tasks could be that SVM can only handle two classes. The approach to handle this issue is to transform the multi-class task into multiple binary tasks and decide the dominant classification with the most votes as its label. Some papers apply SVM to crash severity studies and develop strategies for solving the multi-class problem [8], [18], [41], [42], [43].

C. Ensemble models

Ensemble learning is a paradigm training multiple ML models to deal with the same problem by combining their results. Three major methods (bagging, boosting, and stacking) are used to build meta-algorithms: bagging emphasizes producing a strong learner with less variance, while boosting and stacking train the less-biased ensemble models [44]. The details of ensemble methods are introduced in the following.

1) *Bagging*: In addition to the single classifier for ML, several learning methods that enable aggregating multiple same-type classifiers have attracted attention. Bagging algorithms divide the data into several subsets, and each subset is used for training a single decision tree (at a split node). Then,

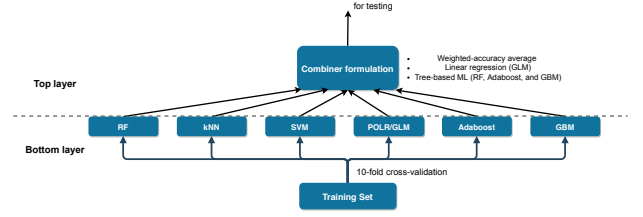


Figure 3: Two-layer stacking machine learning algorithm

the ensemble algorithm integrates the average outcomes of predictions from different decision trees.

Random Forest (RF) extends other tree-based models applying bagging into its learning process, but with randomly distributed independent vectors voting for the majority class [45], [46]. It can be both used to identify the importance of variables and to classify each split node by a random-chosen subset of features. Before the implementation, the number of trees (classifiers) and the number of randomly sampled features at each node should be determined. Some studies employ this method and find it outperforms other ML algorithms [9], [47], [48]. However, its impurity variable importance may be biased towards variables with more classifications or continuous values.

2) *Boosting*: The boosting method is recommended when multiple weak learners (typically of the same pattern) need to aggregate into a strong one. Schapire first proposed this concept, and it has been developed to many well-known algorithms (such as AdaBoost and Gradient Boosting Machine (GBM) for binary classes and Maboost for multi classes) [49], [50], [51]. This method builds trees like RF but allows each classifier to learn from its prior predicted errors. It attempts to reduce the errors iteratively until it reaches the minimum. However, this sequential procedure also has a relatively long computation time and makes the tuning process even harder (considering three parameters number of trees, depth of trees, and learning rate). We exploit the boosting algorithms mentioned above to test if boosting can improve the prediction outcomes.

3) *Stacking*: The concept of the stacking learning technique is to combine different powerful base ML models by training a meta-classifier [52]. No single ML model is perfect. Some models may perform better at some data points, and others predict more accurately at other points. The best combination of their outcomes would use the best model under the appropriate conditions, but this is difficult to achieve, a good combination would weight the better outcomes more heavily. The primary models are trained at the first level, and then a higher-order algorithm is specified at the second level to find the best way for mapping the predictions of bottom-layer learners. The higher-order model (also called the combiner) can be a simple average-weighted formulation (shown in Equation 5), a logistic regression (GLM for binary classification, POLR for multiple classifications), or more sophisticated tree-based models (bagging and boosting) in practice. Figure 3 indicates how stacking works in a two-layer procedure.

$$O_{overall} = \sum_{m=1}^M \alpha_m O_m = \alpha_1 O_1 + \alpha_2 O_2 + \dots + \alpha_M O_M$$

$$\alpha_m = \frac{A_m}{\sum_{m=1}^M A_m} \quad (5)$$

where:

α is the normalized weight depending on the accuracy;

A is the summation accuracy of the base machine learning algorithms. m is the unique ID of different machine learning algorithms (from 1 to M);

O_m is the individual prediction outcomes of base models;

$O_{overall}$ is the overall prediction outcomes of the combiner;

The main achievement of this technique is to integrate the advantages of different models and then improve the accuracy of predictions. Compared to single models, we combine some base models and check whether the combined (ensemble) models perform better or not. Note that to meet the principle of diversity, we also conduct a correlation test for prediction outcomes of ML models.

4) *Ensemble sensitivity analysis*: The demonstration of importance for algorithms requires a sensitivity analysis process. As mentioned before, we cannot directly explain the learning process of classifiers in the ‘black box’. In case that the over-fitting issue occurs in this procedure, the sensitivity analysis is conducted to minimize this effect as well as test the algorithm robustness under uncertainty [53]. One of the conventional approaches of sensitivity analysis is moving input variables one-at-a-time and checking the impact of the rest of the input variables on the performance (one-vs-one method) [54]. The results indicate the importance or the contribution of each input variable to the final accuracy.

To measure the importance of each variable through different underlying algorithms, we first adopt the accuracy-weighted (or information-fusion based) method to ensemble all the outcomes of different algorithms at the top layer [41]. That means the model with the highest accuracy will have the largest weight in explaining the importance of different variables.

Besides a simple accuracy-weighted average formulation, we can also apply a linear (logistic) regression or other complex ML models as combiners at the top layer. The choice of combiners in importance analysis will finally depend on comparison results.

D. Model evaluation metrics

There are several performance assessment methods for ML algorithms. The combination of various evaluation tools is expected to promote the development of analytical research [55]. Therefore, in this study, four fundamental metrics (Accuracy, Sensitivity, Specificity, Area Under the Receiver Operating Characteristics (ROC) Curve (AUC)) will be examined for different algorithms [56], [57], [58]. AUC measures the model’s capability of distinguishing among classes. For example, to evaluate the model performance, if there are only two classifications in the response, the metrics can be expressed in Equation 6.

	Predicted: 1	Predicted: 0
Actual: 1	TP	FP
Actual: 0	FN	TN

Table II: Confusion matrix of two classification predictions. True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN).

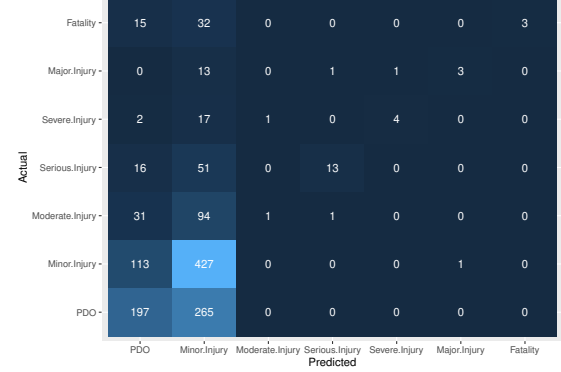


Figure 4: Stacked RF model multi-class confusion matrix learning from unbalance datasets

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (6)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}}$$

$$\text{AUC} = \text{Area under the ROC}$$

True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) (are shown in Table II). As the primary measure of performance, the overall accuracy estimates the ratio of total cases that correctly get classified, but it equally weights false positives and false negatives. The values of sensitivity, specificity, and the area under the receiver operating characteristic possibility curve (AUC) can evaluate how accurately and distinctively the models predict respective positive and negative classes.

IV. RESULTS AND DISCUSSION

A. Multi-class injury severity learning

Table IV summarizes the overall predictive outcomes of four algorithms explicitly. The results indicate, for single models, that RF has the highest prediction accuracy (45.20%) among all models. POLR takes the second place of four models, with the highest AUC value. SVM, kNN, and Maboost show a slightly lower accuracy compared with the other two. However, all models maintain the value of AUC at a relatively low level around 0.6, which is less than desired for the requirement of prediction capabilities. Additionally, stacking slightly improves performance, except for the model stacked by accuracy-weighted average rules. Figure 4 illustrates the stacked RF model confusion matrix.

Bi-class	RF	kNN	SVM	GLM	AdaBoost	GBM
Original	0.7525	0.6081	0.6115	0.6235	0.7323	0.7564
Oversampling	0.7264	0.5903	0.6034	0.6341	0.7398	0.7669
Undersampling	0.6562	0.6404	0.6441	0.6095	0.6223	0.7585
SMOTE	0.7271	0.6865	0.7297	0.7221	0.6887	0.7832
ROSE	0.6451	0.6443	0.6259	0.6202	0.6323	0.6269
Tri-class	RF	kNN	SVM	POLR	Maboost	
Original	0.6740	0.5753	0.6223	0.6108	0.6904	
Oversampling	0.6736	0.5661	0.6122	0.6061	0.6915	
Undersampling	0.6735	0.5846	0.6103	0.6087	0.6979	
SMOTE	0.6202	0.5574	0.5857	0.5910	0.6790	

Table III: AUC value results of different aggregation types and sampling methods

Model Name	Tuning Parameters	Train Accuracy	Test Accuracy	95% CI	Sensitivity	Specificity	AUC	Aggregation Accuracy
RF	mtry = 2	43.79%	45.20%	(42.84%, 47.58%)	0.7199	0.4892	0.6102	86.94%
kNN	k = 13	40.15%	41.27%	(38.94%, 43.63%)	0.7000	0.3580	0.5825	87.33%
SVM	C = 0.25 & sigma = 0.035	43.10%	44.97%	(42.61%, 47.35%)	0.7236	0.4024	0.5309	87.40%
POLR	method = cauchit	44.85%	45.03%	(42.67%, 47.41%)	0.7379	0.4191	0.6161	87.63%
Maboost	Method: normal, Iteration: 100	45.70%	43.86%	(41.14%, 46.60%)	0.7203	0.4398	0.5566	86.79%
Stacking with accuracy-weighted formula		44.12%	41.74%	(41.74%, 41.74%)	0.6443	0.3128	0.5000	86.79%
Stacking with RF	mtry = 16	46.47%	49.77%	(47.02%, 52.52%)	0.7666	0.5508	0.6002	88.56%
Stacking with POLR	method = cauchit	48.69%	49.30%	(46.27%, 50.04%)	0.7392	0.4437	0.6184	88.81%
Stacking with Maboost	Method: normal, Iteration: 100	47.10%	47.08%	(44.34%, 49.84%)	0.7462	0.4688	0.6200	87.56%

Table IV: Prediction performance of different multi-class ML algorithms

Model Name	Tuning Parameters	Train Accuracy	Test Accuracy	95% CI	Sensitivity	Specificity	AUC
RF	mtry = 2	85.04%	84.72%	(82.64%, 86.63%)	0.9016	0.4056	0.7271
kNN	k=13	72.51%	70.58%	(68.03%, 73.05%)	0.9100	0.2302	0.6865
SVM	C = 0.25 & sigma = 0.035	78.29%	78.88%	(76.56%, 81.07%)	0.9171	0.3141	0.7297
GLM		80.57%	81.11%	(78.87%, 83.20%)	0.9100	0.3348	0.7221
GBM	n.trees = 150, interaction.depth = 3 & shrinkage = 0.1	84.20%	85.18%	(83.13%, 87.06%)	0.9056	0.4286	0.7132
Adaboost	nIter = 150 & method = RealAda	82.37%	82.80%	(80.63%, 84.81%)	0.9074	0.3632	0.6887
Stacking with accuracy-weighted formula		80.85%	73.89%	(71.41%, 76.25%)	0.9047	0.2423	0.5735
Stacking with RF	mtry = 2	88.19%	88.02%	(86.13%, 89.73%)	0.8902	0.6481	0.6873
Stacking with GLM		87.37%	88.40%	(86.54%, 90.09%)	0.8913	0.7059	0.7226
Stacking with GBM	n.trees = 150, interaction.depth = 3 & shrinkage = 0.1	87.81%	87.94%	(86.05%, 89.66%)	0.8864	0.6744	0.7327
Stacking with Adaboost	nIter = 150 & method = RealAda	87.93%	87.56%	(85.64%, 89.30%)	0.8884	0.5893	0.6022

Table V: Prediction performance of different bi-class ML algorithms

The problem with multiple classifications of injury severity is that the minority (in this case, severe injuries and fatalities) takes quite small proportions in this dataset, such as Level 4 (2.15%) and Level 5 (1.10%). This imbalance indicates that multi-class models tend to improve their overall accuracy at the cost of ignoring small categories, which leads to a relatively low AUC (around 0.6). It is also the main reason for the limited improvements of ensemble multi-class models. Nevertheless, we find the aggregation accuracy of multi-class models exceeds the default proportion of larger categories classes (86.52%) (if all predictions were for the largest categories).

To address this issue, we aggregate classifications into two categories [59], [60], [8]. In this study, we divide seven

classifications of the injury severity by Level 3 into two groups (bi-class labels ‘3-’ and ‘3 & 3+’; tri-class labels ‘0’, ‘1’, and ‘2 & 2+’). The frequency of aggregated classifications is shown in Table I. After that, we resample the ‘Training’ set inside of the cross-validation process by different methods (no sampling, oversampling, undersampling, SMOTE, and ROSE (only valid in bi-class models)) and check if the bias problem can be resolved. **Details of these methods can be found in [61], [62], [63].** The AUC value (that measures the model performance with imbalanced classifications) results are shown in Table III. We finally choose the two-category aggregation with SMOTE because of its highest predictive power in more models than other combinations.

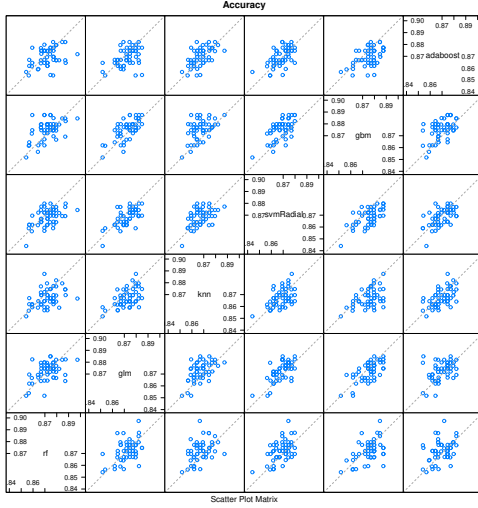


Figure 5: Correlation of underlying models of the stacking technique

B. Bi-class injury severity learning

We deploy the same ML models with the multi-class learning process for binary classifications. The prediction results are shown in Table V. Most primary models can make predictions with an accuracy $> 80\%$ and an AUC > 0.7 . The accuracy of multi-class models aggregated into two categories is better than bi-class models, recognizing it overemphasizes the larger categories at the expense of the smaller categories, which results in imbalanced learning. Aggregation and subsampling approaches before training may mitigate the problem to some extent.

Boosting and stacking techniques can also be deployed in bi-class learners. The results indicate that the better bi-class stacking models can significantly improve the specificity and yield prediction performance better over any single trained models.

Among stacking models, the GLM combiner makes better predictions than the accuracy-weighted and tree-based combiners, which indicates a linear blender is more suitable for this ensemble bucket.

Then, instead of training the bottom-layer models by the same single training and testing sets in the stacking process, we employ more randomly-chosen splits for testing subsets in each underlying model. The results in Table VI indicate that, in most cases, one split is sufficient for a robust (within the narrow range of confidence intervals) and accurate prediction. Splitting into more subsets does not demonstrate distinctive advantages over fewer subsets but adds some extra complexity. This depends on how data is subdivided in small or medium-size datasets.

To ensure the effectiveness of stacking models, we further check the correlation of prediction outcomes for underlying models (presented in Figure 5). There are no coefficients that are more than 0.75 in the correlation table, which means the underlying models are sufficiently independent for making predictions.

Bagging (RF) and boosting algorithms (GBM and Ad-

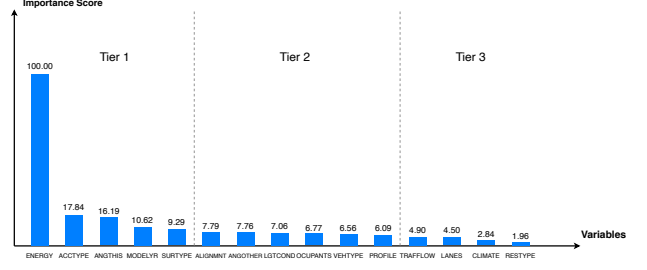


Figure 6: Importance score for various features ensemble by GLM (scaled to 100)

aBoost) also perform well by iteratively enhancing the classifiers.

Overall, the model results generally have good sensitivities, fair AUC values, and unsatisfactory specificities, which are comparable to most outcomes in other similar studies [64], [65], [41], [66]. The differences are caused by various injury classification criteria, independent variable sets, and the total number of cases.

C. Overall variable importance analysis of ML algorithms

An ensemble sensitivity analysis with the bi-class dependent variable is conducted to identify the variable importance in Figure 6. Given that the GLM combiner outperforms any other combiners in the results, we apply its ensemble weighting to this importance analysis. It is found that energy absorption dominates in the importance score list, which contributes to increasing the prediction accuracy. The top group (Tier 1), such as crash type and collision angle, exceeds a score of 9 in the chart. Features in Tier 2 present less significant prediction capability than Tier 1, ranging from 5 to 9. The remaining features in Tier 3 also affect the outcomes but with limited influence.

Energy absorption is demonstrated as one of the essential components that can dramatically raise the accuracy of ML models. That is because it has a direct impact on the outcomes of occupants' injury severity with low correlations with other features. However, the specific influence (whether proportional or inversely proportional to injury severity and to what extent) of energy absorption should be handled by a statistical analysis [28]. As another crash mechanism-related predictor, the collision angles also greatly affect the prediction process.

Other significant features like crash types, surface types, and light conditions, which also have a great impact on the injury severity, are consistent with conclusions from previous studies [67], [24], [68], [69], [41]. However, the remaining factors may also have less predictive power. Even though some studies suggested some less-effective variables based on this ranking (restraint type, weather condition, etc.) are important, our study does not observe these advantages because of the significance of the other features. Following this importance ranking, top features in this analysis can be valued in not only statistical analysis but outcome prediction for future injury severity studies.

Multi-class	Accuracy-weighted formula	95% CI	Stacked with RF	95% CI	Stacked with POLR	95% CI	Stacked with Maboost	95% CI
One Split	41.74%	(41.74%, 41.74%)	49.77%	(47.02%, 52.52%)	49.30%	(46.27%, 50.04%)	47.08%	(44.34%, 49.84%)
Two Splits	42.86%	(38.15%, 47.66%)	39.40%	(34.77%, 44.17%)	38.52%	(32.54%, 42.05%)	42.40%	(37.70%, 47.20%)
Three Splits	39.08%	(33.74%, 44.62%)	37.54%	(32.26%, 43.05%)	40.97%	(34.63%, 45.51%)	40.00%	(34.63%, 45.55%)
Four Splits	45.00%	(38.85%, 51.27%)	39.23%	(33.26%, 45.45%)	39.41%	(33.37%, 45.69%)	37.31%	(31.41%, 43.50%)
Bi-class	Accuracy-weighted formula	95% CI	Stacked with RF	95% CI	Stacked with GLM	95% CI	Stacked with GBM	95% CI
One Split	73.89%	(71.41%, 76.25%)	88.02%	(86.13%, 89.73%)	88.40%	(86.54%, 90.09%)	87.94%	(86.05%, 89.66%)
Two Splits	65.21%	(60.52%, 69.69%)	85.94%	(82.31%, 89.08%)	85.53%	(82.19%, 88.97%)	86.18%	(82.57%, 89.28%)
Three Splits	59.08%	(53.51%, 64.47%)	87.69%	(83.62%, 91.06%)	88.00%	(83.45%, 91.20%)	86.29%	(80.36%, 88.24%)
Four Splits	59.23%	(52.99%, 65.26%)	88.08%	(83.50%, 91.75%)	88.36%	(83.84%, 91.98%)	88.46%	(84.02%, 92.16%)

Table VI: The results of different subset splits in the stacking process. **Bold** indicates best performance.

V. CONCLUSIONS

Due to the difficulty and complexity of the prediction for injury severity in crashes, machine learning algorithms are attractive for crash analysis because of their strong learning abilities. In contrast with other studies, we highlight and compare the influence of crash mechanism-related features. This paper adopts several popular ML models and some ensemble techniques to predict occupants' injuries in two-vehicle crashes. The NASS/CDS datasets were applied in this study, but the information provided in the current datasets is incomplete, (in particular, police reports may miss many unreported incidents). Future large datasets with fewer unknown items should enable ML algorithms to improve AUC and specificity over current results.

The prediction results indicate that aggregation and resampling can reduce the bias problem associated with a multi-class dependent variable. For single models, the GBM and the RF perform the best in accuracy with their iterative improving algorithms. Combining models using a stacking approach can improve the performance over base models, which is consistent with our hypothesis. To mitigate the 'black box' effect, we introduce the ensemble sensitivity analysis to avoid over-fitting and further check the impact of various features. After ranking their importance, we find that energy absorption contributes the most to the prediction accuracy, and other variables also have significant effects on the outcomes.

Future research should aim to include more features into feature selection with the support of more detailed data. Future research can also extend the scope of two-vehicle to multiple-vehicle involved crashes. That requires further exploration of multi-vehicle crash mechanisms, and a larger dataset, as multi-vehicle crashes are rarer events.

NOMENCLATURE

α	the normalized weight based on the accuracy
β, β_0	the corresponding coefficient of variable X and the intercept
$\bar{a}$	the average acceleration imposed on vehicles
$\bar{m}$	the mass at the gravity center of the uncrashed vehicle parts
ε	the error term in the regression function
A	the summation accuracy of the base models
d	the crash displacement
E	the energy absorption
M	the unique ID of different machine learning algorithms (from 1 to M)

m	the number of classifications for the dependent variable
N	the total number of independent variables
O_m	the individual prediction outcomes of base models
$O_{overall}$	the overall prediction outcomes of the combiner
s, p	The subject vehicle and the partner vehicle
AUC	Area Under the ROC possibility Curve
CDS	Crashworthiness Data System
CI	Confidence Interval
GBM	Gradient Boost Machine
GLM	Generalized Linear Model
kNN	k-Nearest Neighbor
MAIS	Maximum Abbreviated Injury Score
MICE	Multiple Imputations by Chain Equations
ML	Machine Learning
NASS	National Automotive Sampling System
NHTSA	National Highway Traffic Safety Administration
OLR	Ordered Logistic Regression
PMM	Predictive Mean Matching
POLR	Proportional odds Ordered Logistic Regression
RF	Random Forest
RFE	Recursive Feature Elimination
ROC	Receiver Operating Characteristic
ROSE	Random Over-Sampling Examples
SMOTE	Synthetic Minority Over-sampling TEchnique
SVM	Support Vector Machines

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